

Website: salmanblogs.com | Youtube: ComputerWithSalman

These lecture notes are prepared by **Salman Ahmad**, who is both a **teacher** and a **web developer**. The purpose of this material is to make the topic clear and useful for students of Class 9 Computer Science.

Algorithm Design Methods [Lecture # 2]

Algorithm design methods are the different ways or techniques used to create and plan an algorithm (step-by-step solution to a problem).

There are 2 Methods to design an algorithm:

1. Flowcharts
2. Pseudocode

Flowcharts

A flowchart is a diagram that shows the steps of a process to solve any problem using symbols and arrows. It shows how a problem is solved in a visual way.

Used in: Computer Science, Engineering, Business, and many fields of life.

Importance of Flowcharts

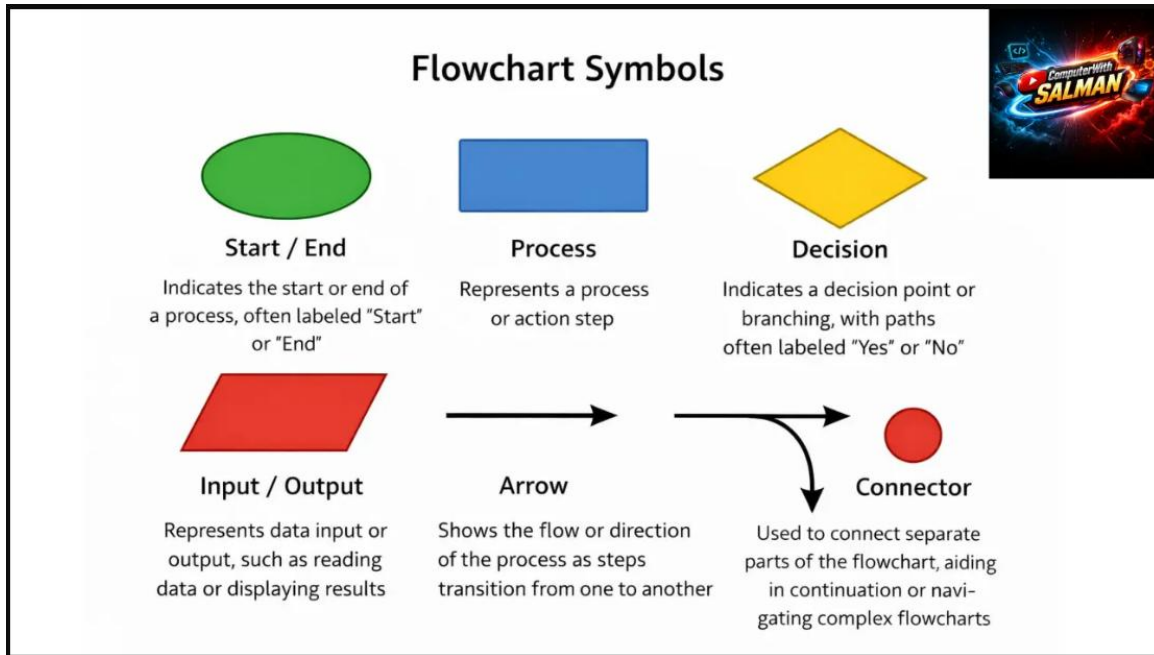
Clarity: Flowcharts present processes in a clear and organized way, making them easy to understand quickly.

Communication: They help explain complex ideas in a simple visual form so everyone can understand easily.

Problem Solving: Flowcharts help in understanding the steps of a problem clearly in pictorial form, making it easier to think of better and more effective solutions.

Documentation: They are useful for saving process details for learning, training, and future reference.

Flowchart Symbols



Pseudocode

Pseudocode is a way of writing an algorithm using simple English-like statements. It is similar to programming logic but without strict syntax.

Example:

```
START INPUT A
INPUT B
SUM = A + B PRINT
SUM END
```

Why Use Pseudocode?

Clarity: Pseudocode is easy to understand because it focuses only on the logic of the solution without worrying about complex programming rules or syntax.

Planning: It helps programmers organize their ideas and plan the steps of the algorithm before writing real code.

Communication: It provides a simple way to explain algorithms to others, so everyone can understand the logic clearly, even without programming knowledge.

Flowcharts vs Pseudocode

Pseudocode:

- Written in simple English.
- Looks like a story.
- Steps are written one by one.
- Easy to convert into programming code.

Flowcharts:

- Uses diagrams and symbols.
- Looks like a visual map.
- Shows flow using arrows.
- Easy to understand the overall process quickly.

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